

Theory of Computation

CSIT — 4th Semester — 2079 — Answer Sheet
Subject Code: CSC262

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. Show that, For any NFA $(N = (Q_N, \Sigma, \delta_N, q_0, F_N))$, there exists a DFA $(D = (Q_D, \Sigma, \delta_D, q_0, F_D))$ such that $L(N) = L(D)$ for $(L \subseteq \Sigma^*)$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. State and prove the Pumping Lemma for regular languages. How can you show with example that pumping lemma is used to prove that a given language is not a regular? Explain.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Given the following expression grammar for simple arithmetic expression with operator + and *. Remove left recursions from this grammar then simplify and convert to CNF. $(E \rightarrow E + T \mid T) \mid (T \rightarrow T * F \mid F) \mid (F \rightarrow (E) \mid a)$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions [8×5=40]

4. Explain the (ϵ) -closure of states of an (ϵ) -NFA with a suitable example

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Convert the following regular expression into equivalent Finite Automata. a. $((0 + 1)^* 10(1 + 0)^*)$ b. $(1^* 0(0 + 1)^* 1)$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Define the term: Parse Tree, Left-most and right-most derivation, sentential form and ambiguity with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Give the formal definition of Push Down Automata. How CFG can be converted into equivalent PDA? Explain with an example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Define regular grammar. Also explain the method of converting right linear grammar into equivalent Finite Automata.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Construct a Turing Machine that accept a language $L = \{a^n b^n \mid n > 0\}$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Define Turing Machine and explain its roles.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Explain about the complexity classes P, NP and NP-complete.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Write short notes (Any two) a. Big Oh, Big Omega and Big Theta. b. Tractable and Intractable problems. c. Chomsky Hierarchy.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.